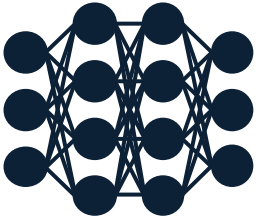


Understanding LSTMs: Long Short-Term Memory Networks

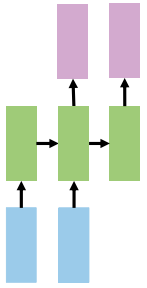
Adam Dziedzic

May 12th, 2026

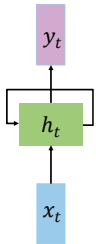
Plan of the Lecture



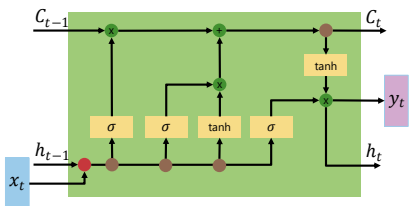
1. Motivation: why standard neural networks insufficient?



2. Operate on sequence of vectors & remember the past

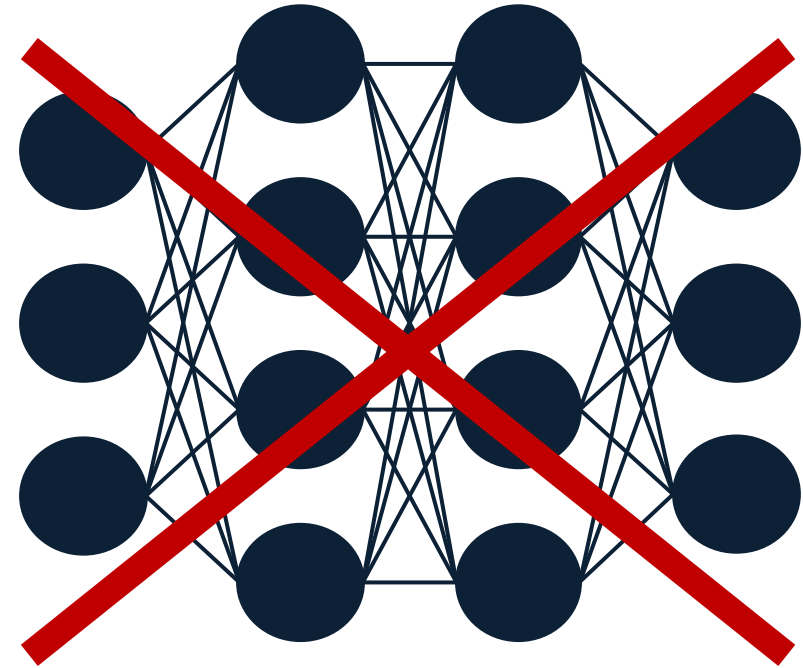


3. RNNs: Recurrent Neural Networks



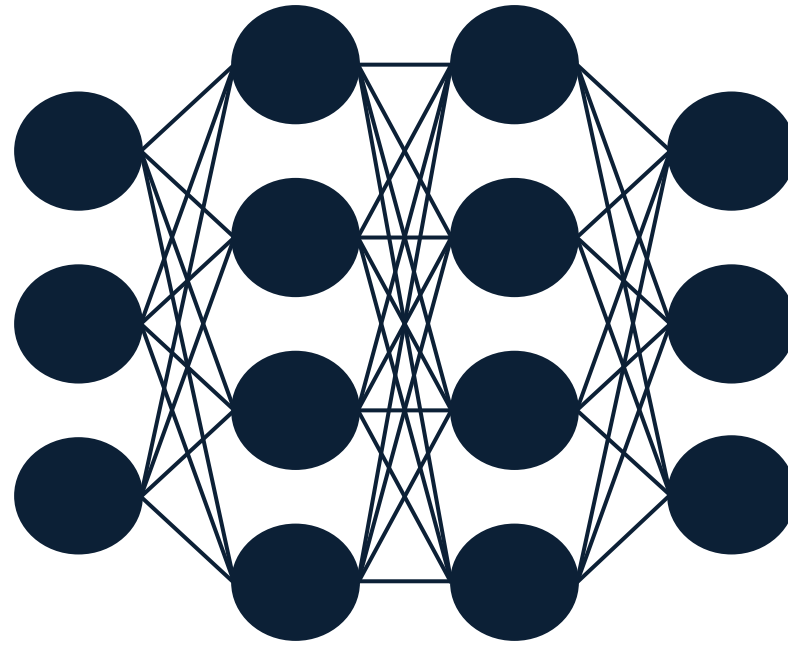
4. LSTMs: Long Short-Term Memory Networks

Importance of Memory in Humans & Networks



No Memory

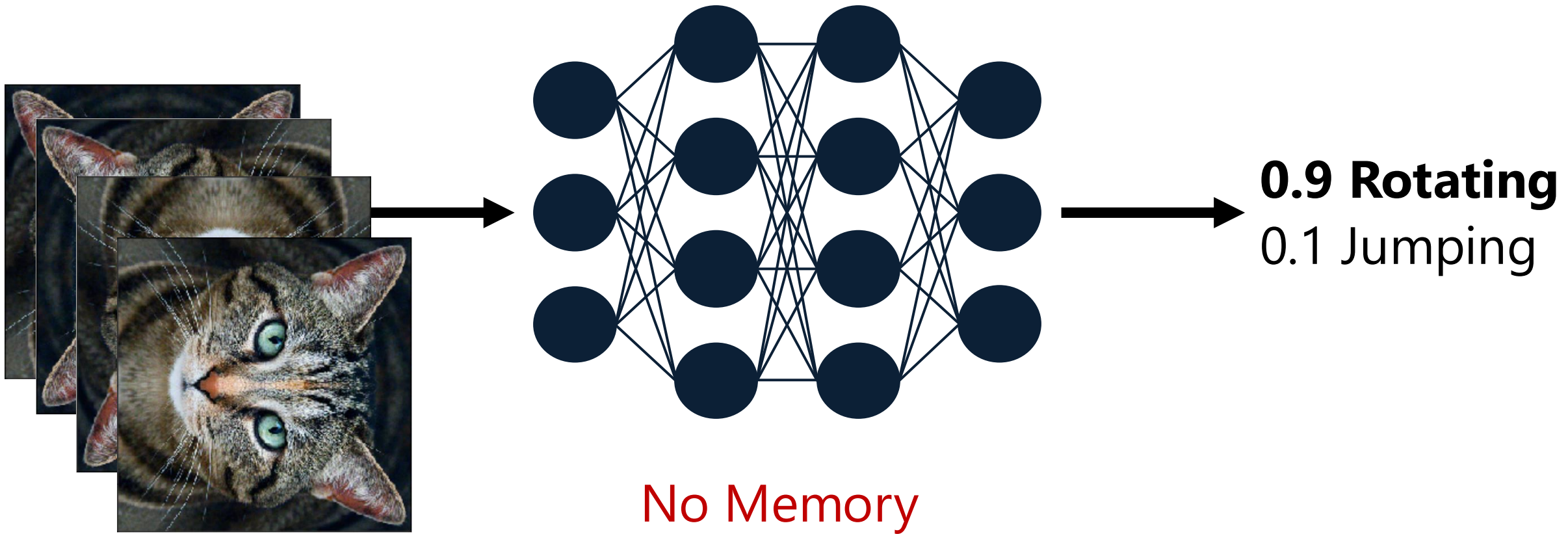
No Memory in Standard Neural Networks



0.9 Cat
0.1 Dog

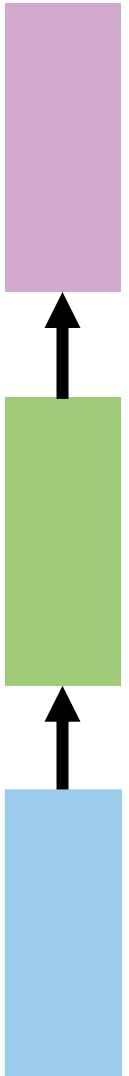
No Memory

No Memory in Standard Neural Networks

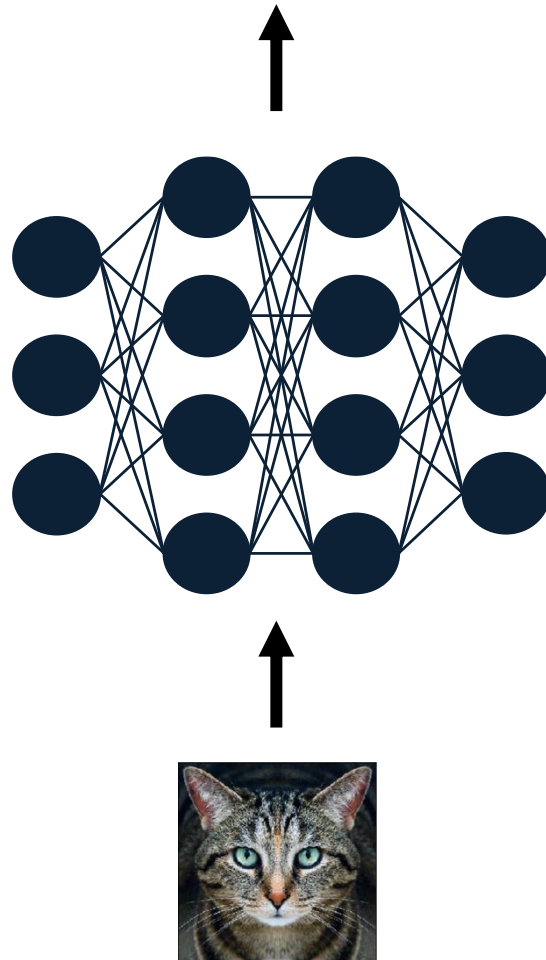


Motivation: Operate On Sequences of Vectors

one to one

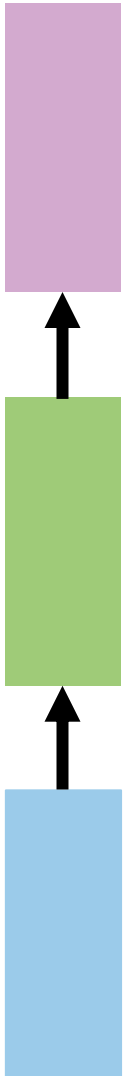


0.9 Cat
0.1 Dog

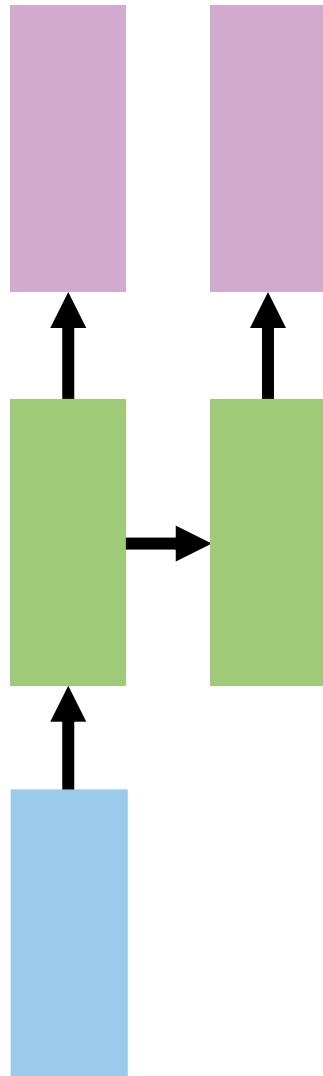


Motivation: Operate On Sequences of Vectors

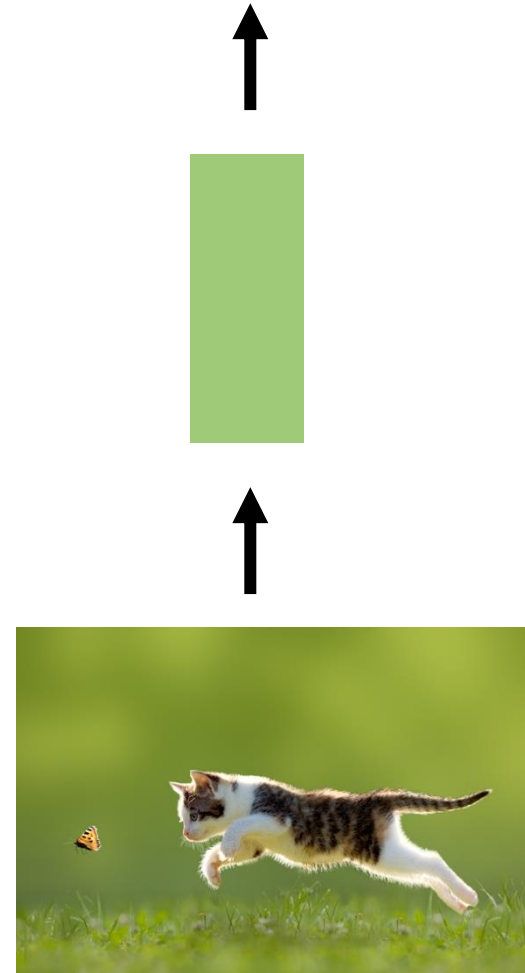
one to one



one to many

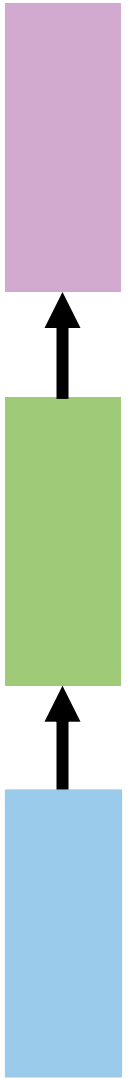


"A cat is chasing a butterfly on a grass."

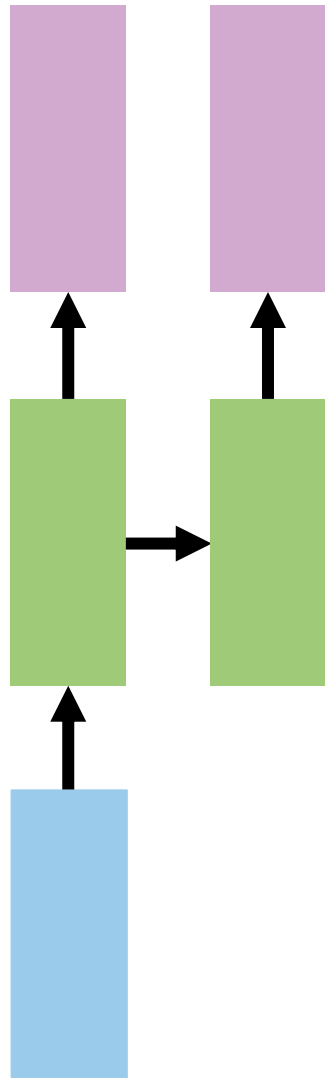


Motivation: Operate On Sequences of Vectors

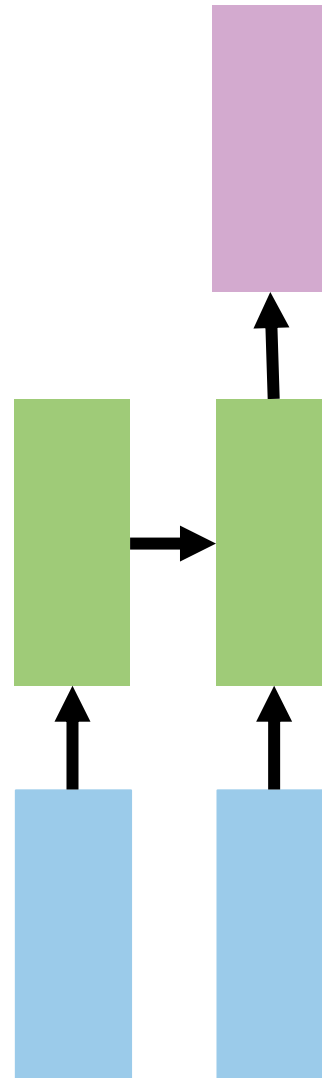
one to one



one to many



many to one



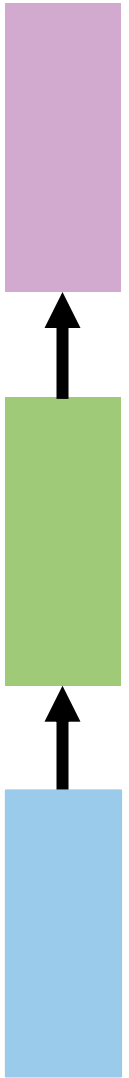
Sentiment: positive



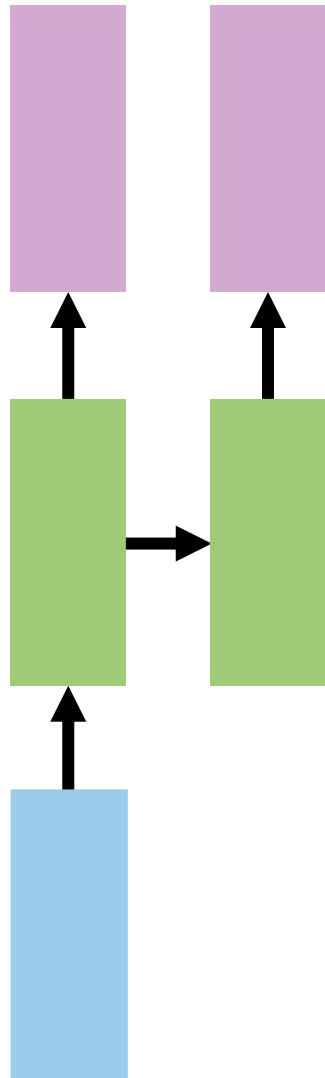
"I would put this at the top of my list of films"

Motivation: Operate On Sequences of Vectors

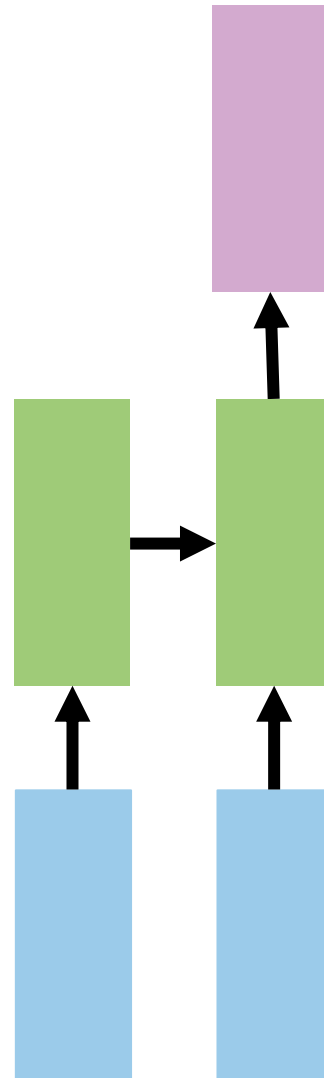
one to one



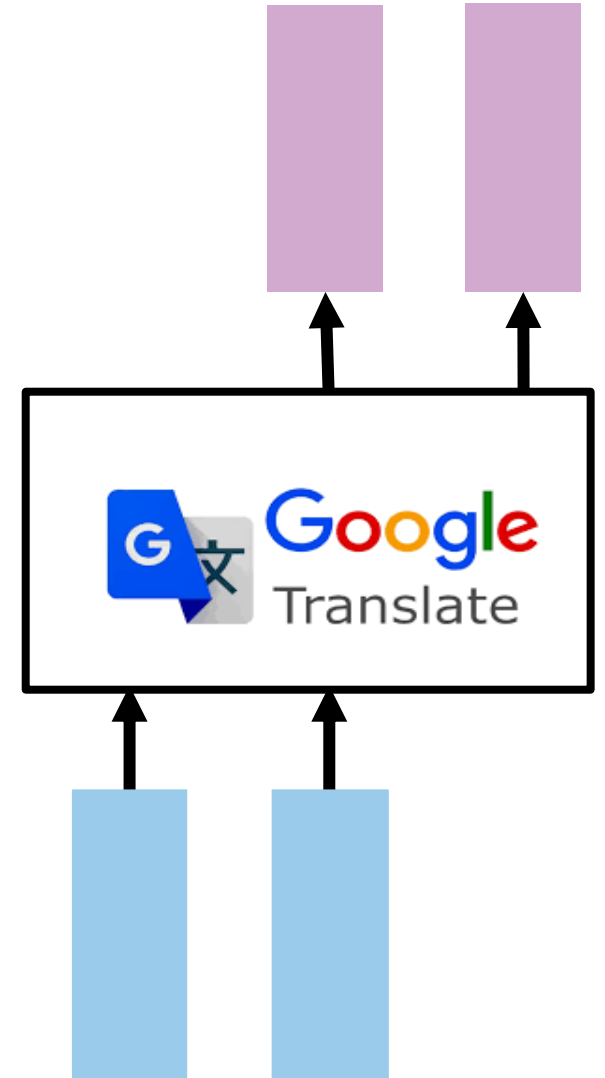
one to many



many to one

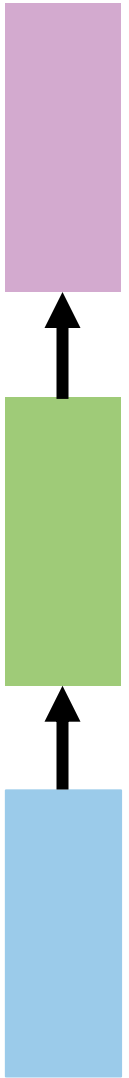


many to many

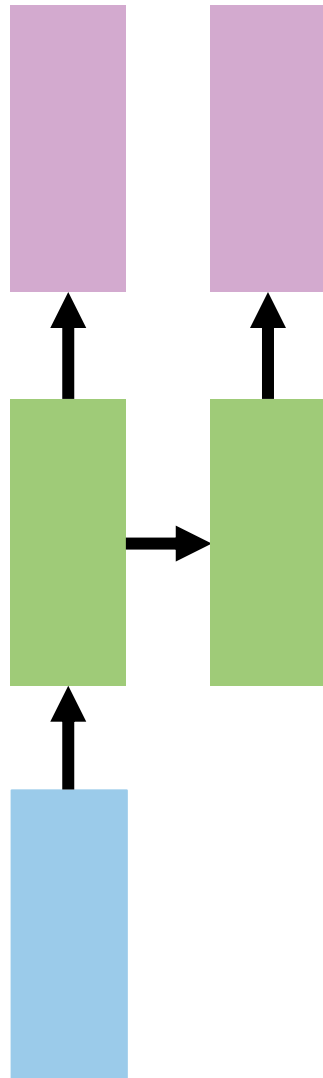


Motivation: Operate On Sequences of Vectors

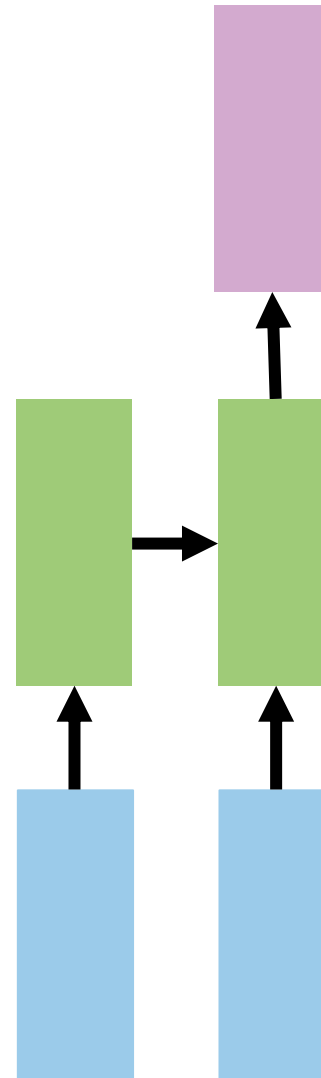
one to one



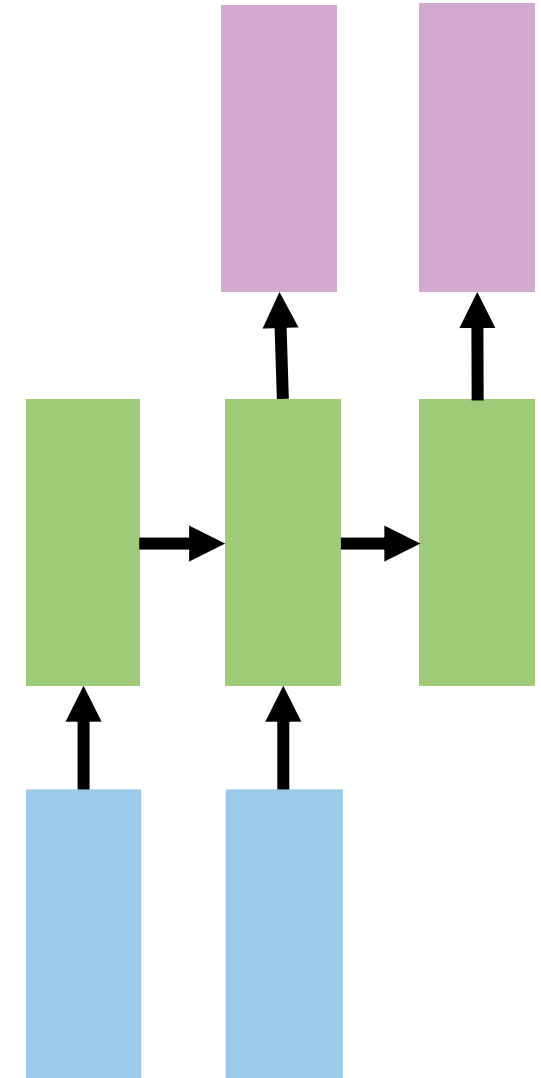
one to many



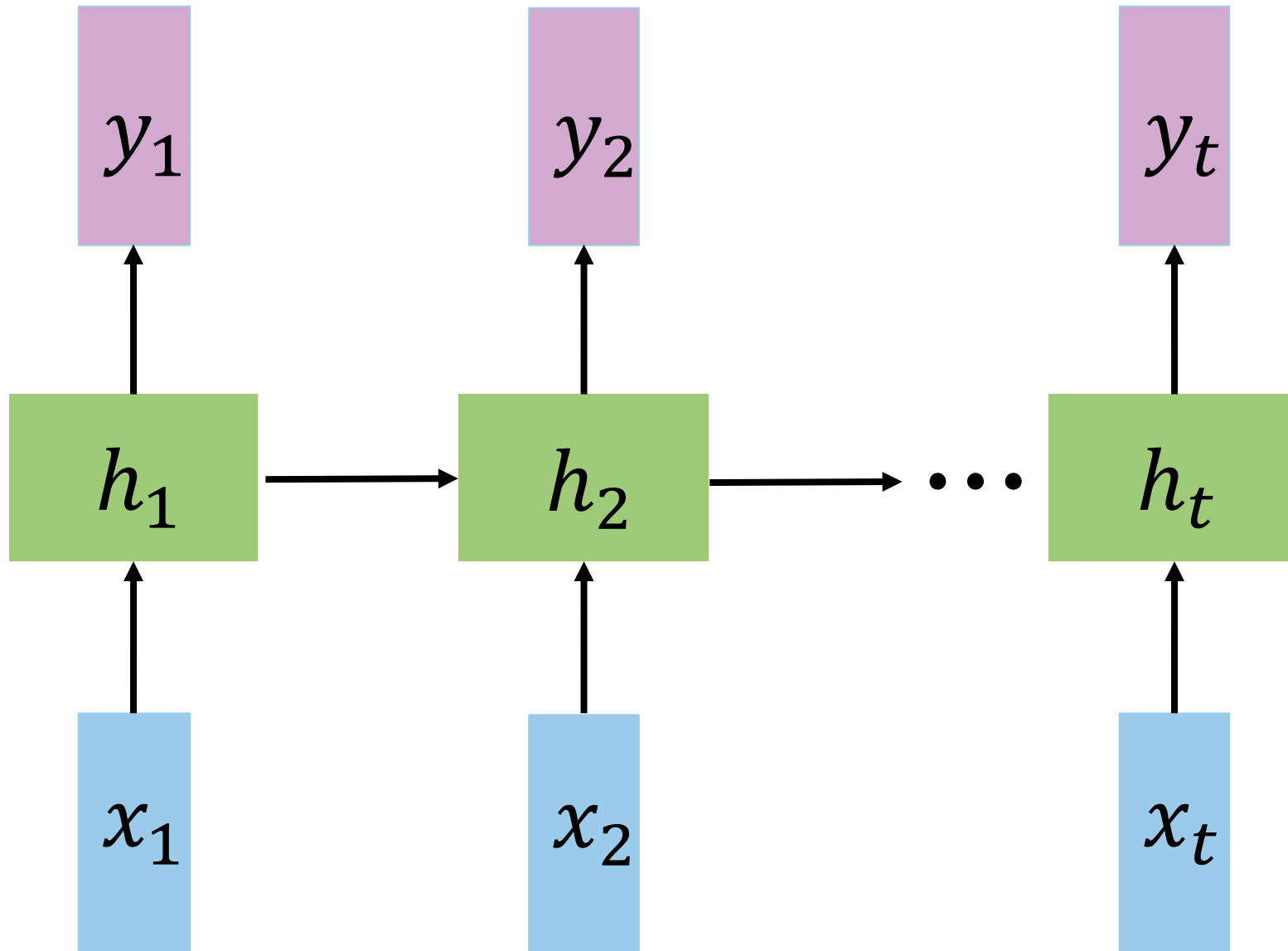
many to one



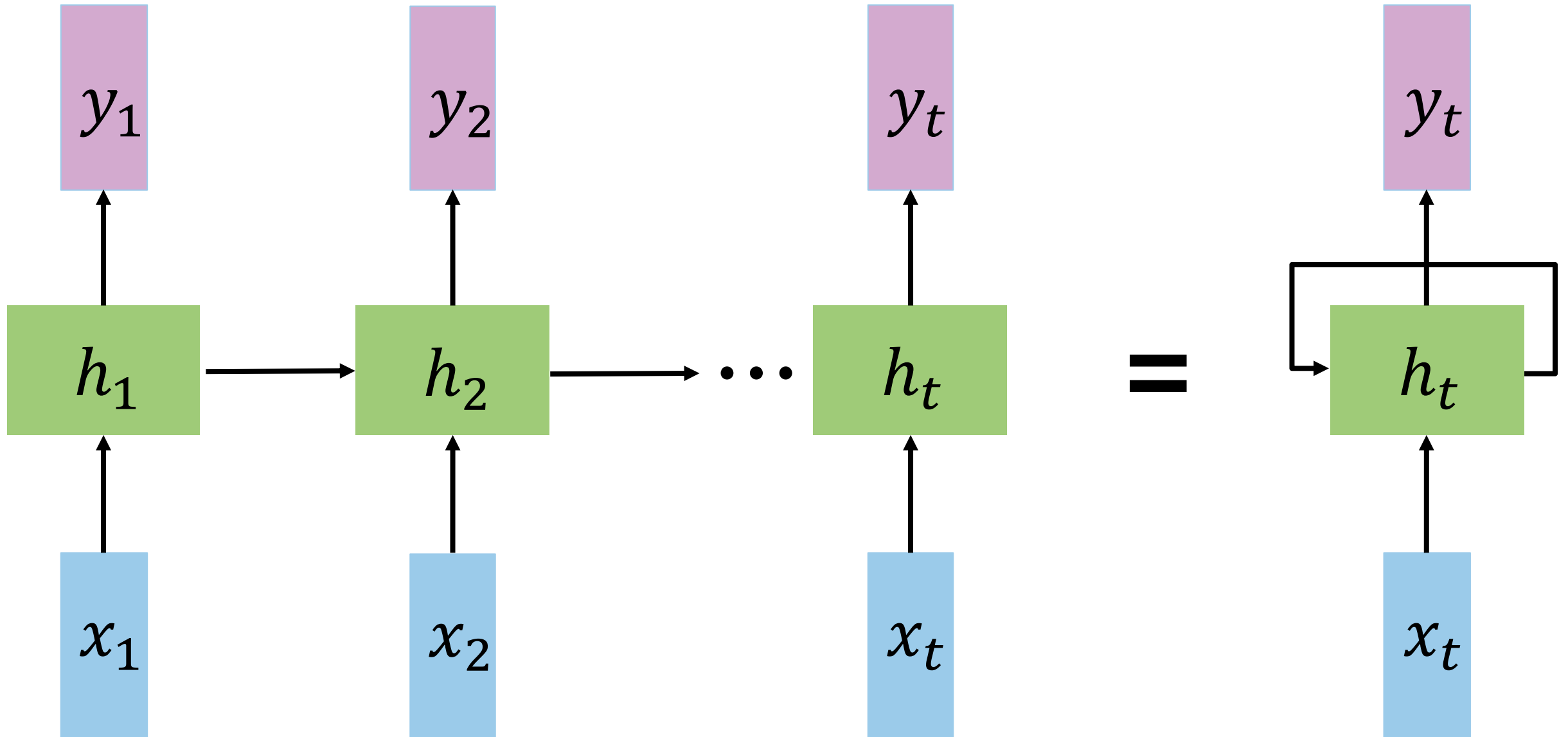
many to many



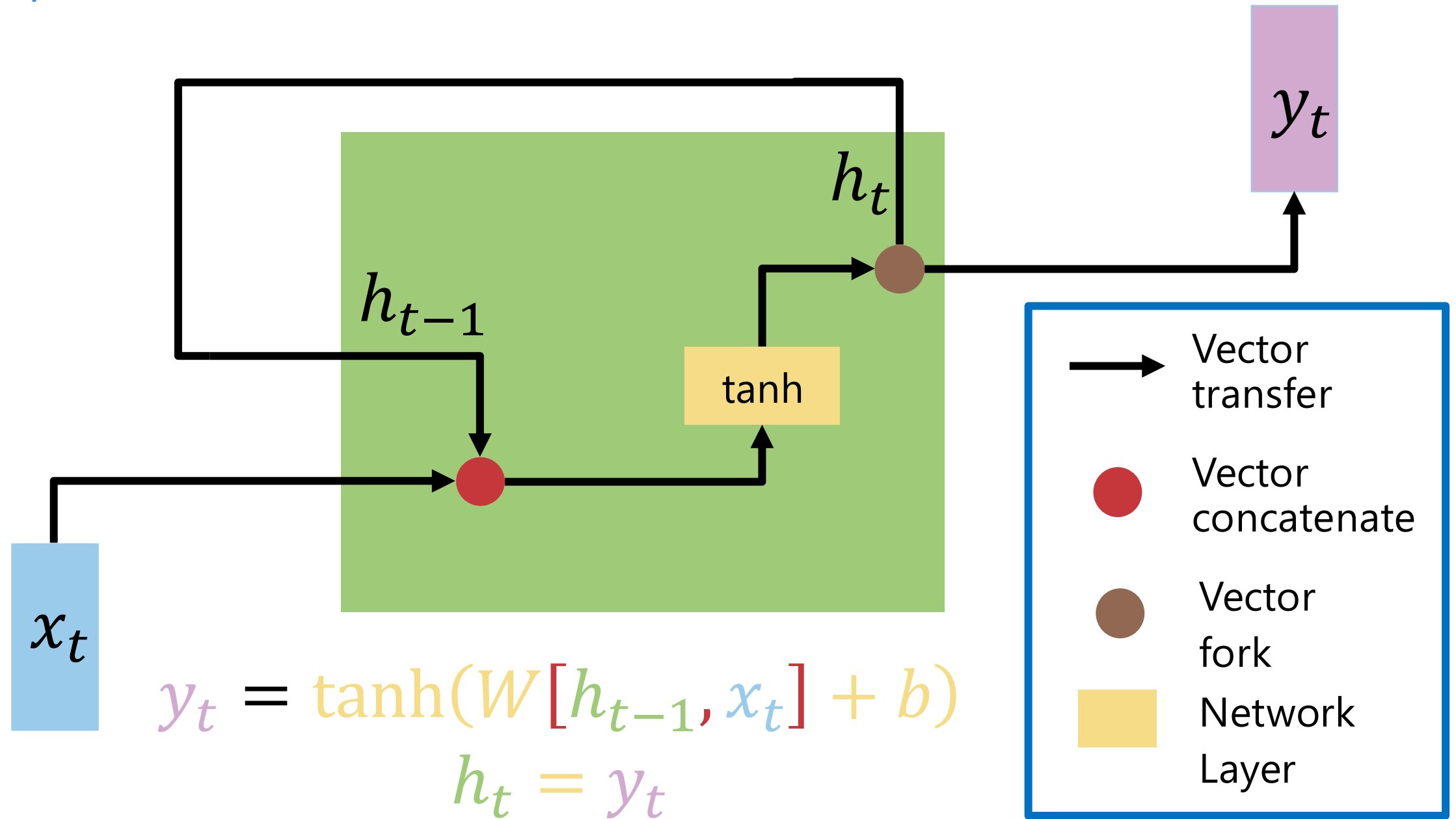
RNNs: Recurrent Neural Networks



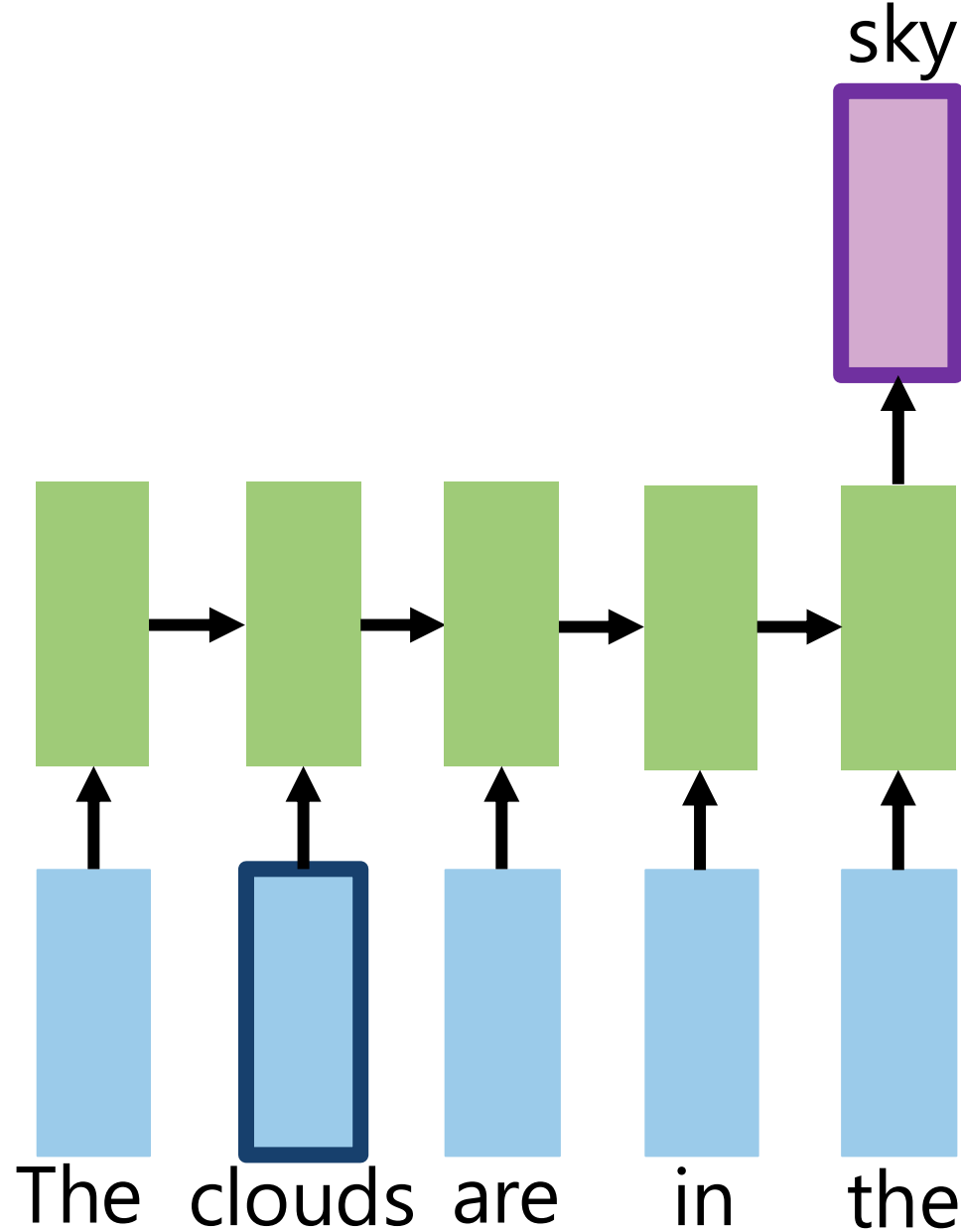
RNNs Remember Sequences of Vectors



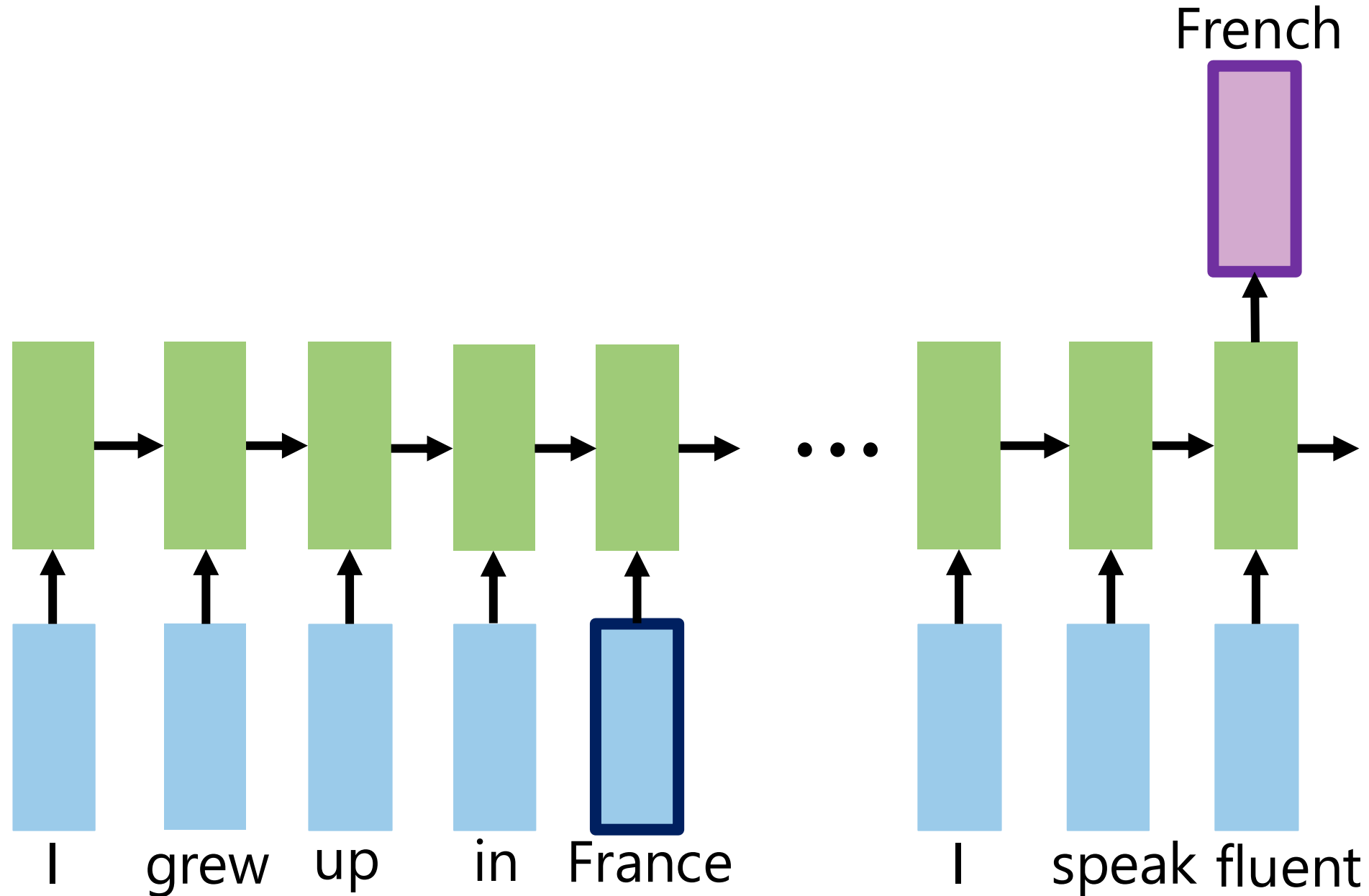
Computation in Recurrent Neural Network Cell



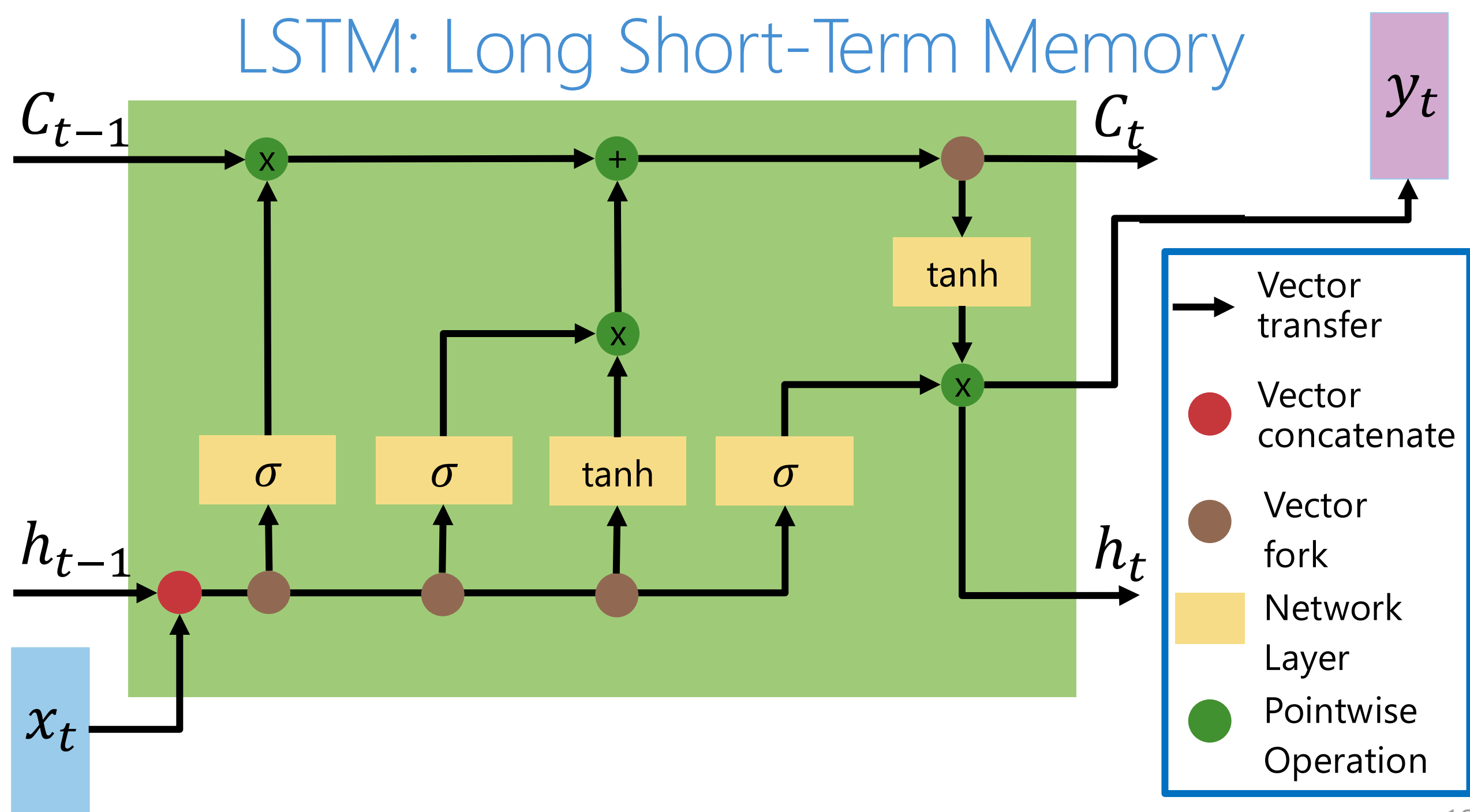
Problem of Long-Term Dependencies



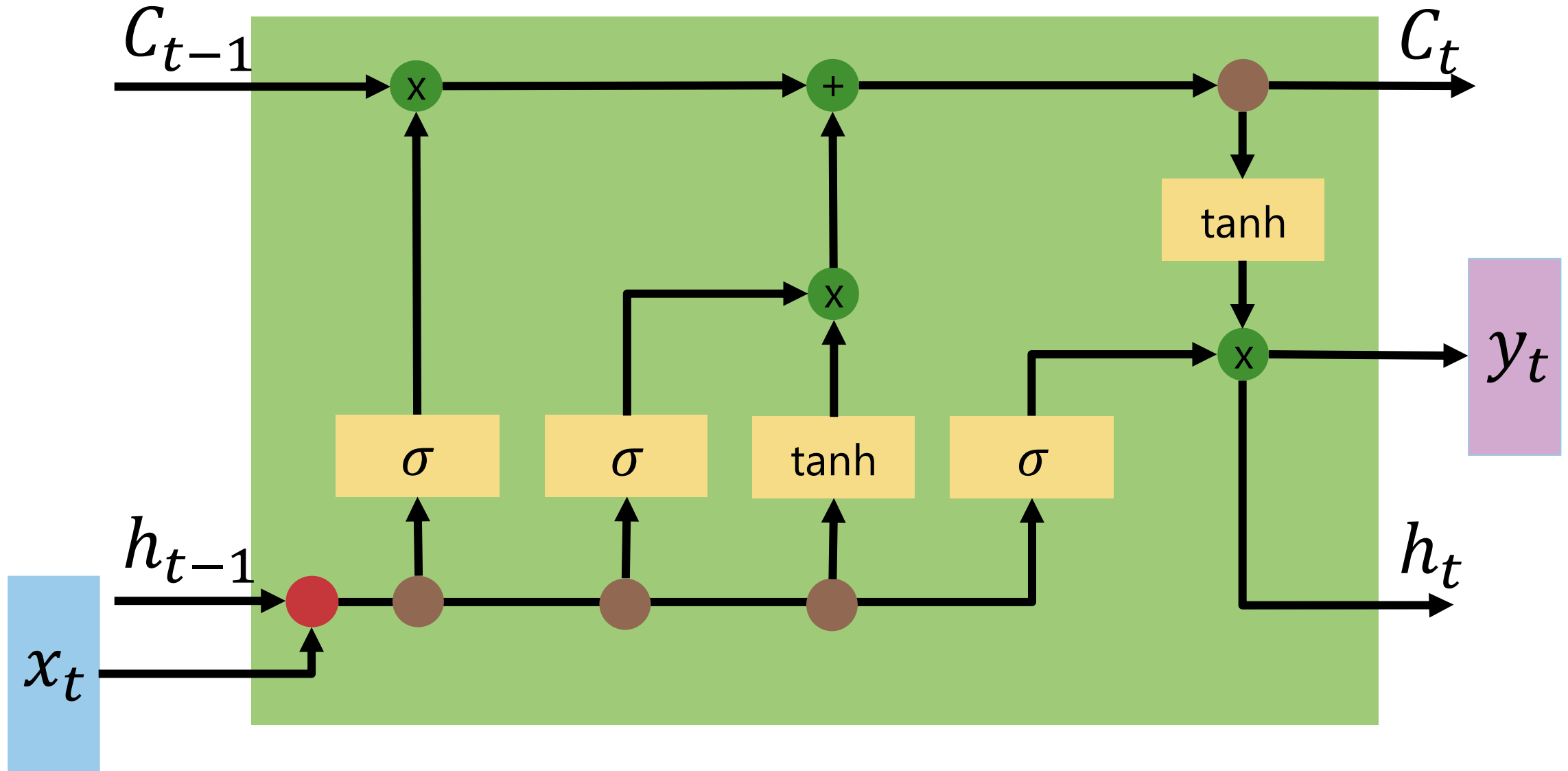
Problem of Long-Term Dependencies



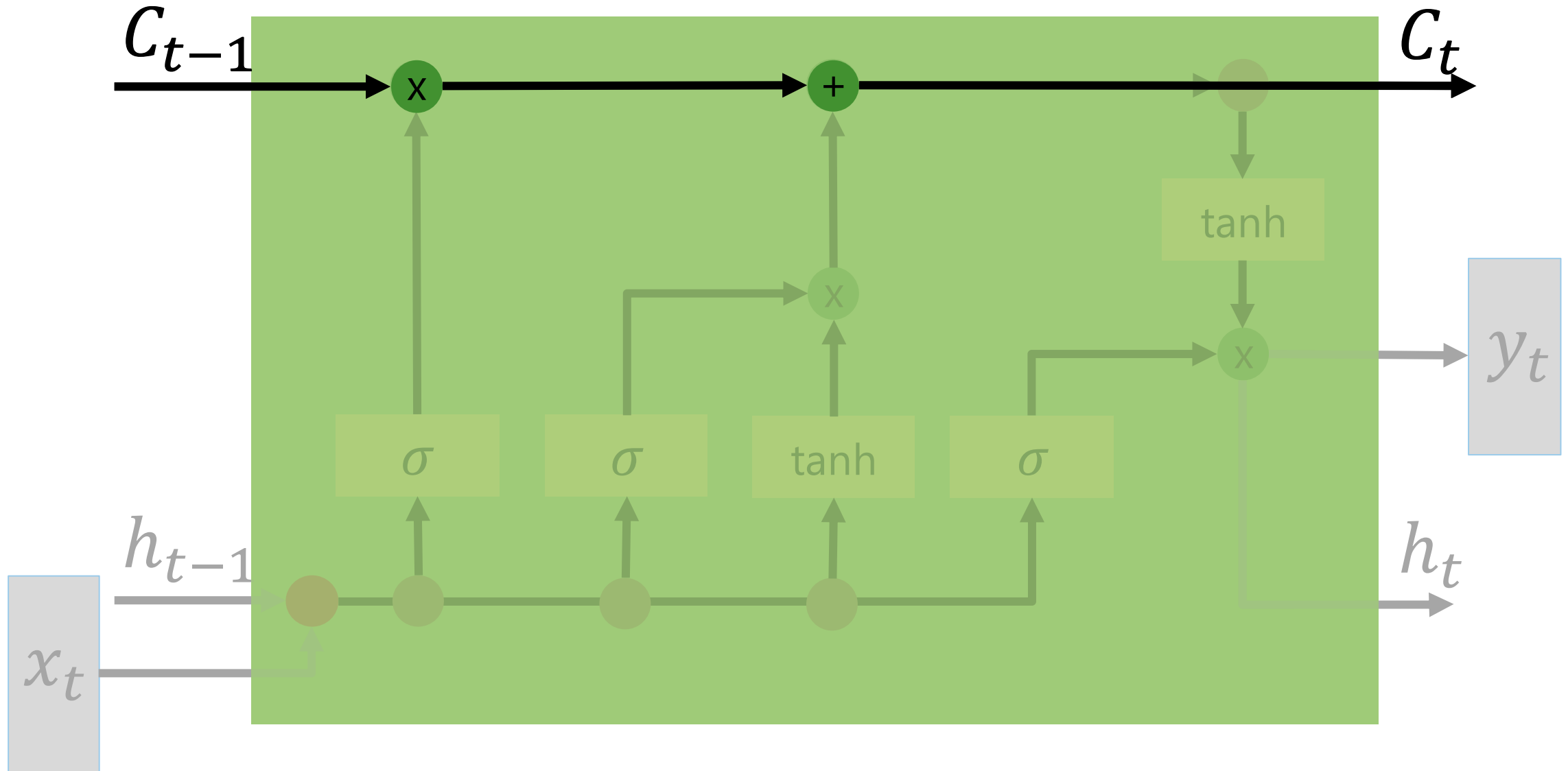
LSTM: Long Short-Term Memory



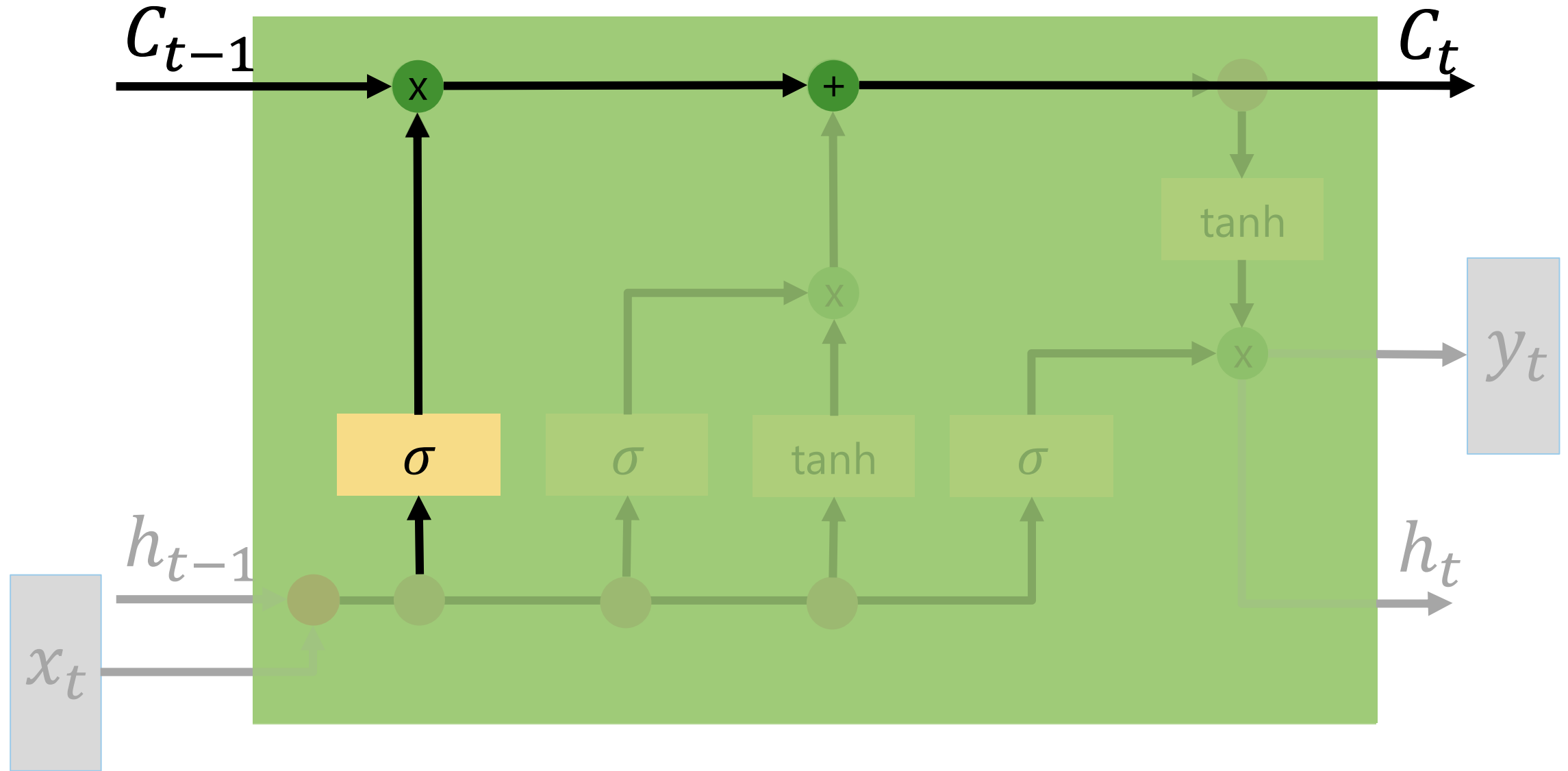
LSTM: Step by Step Explanation



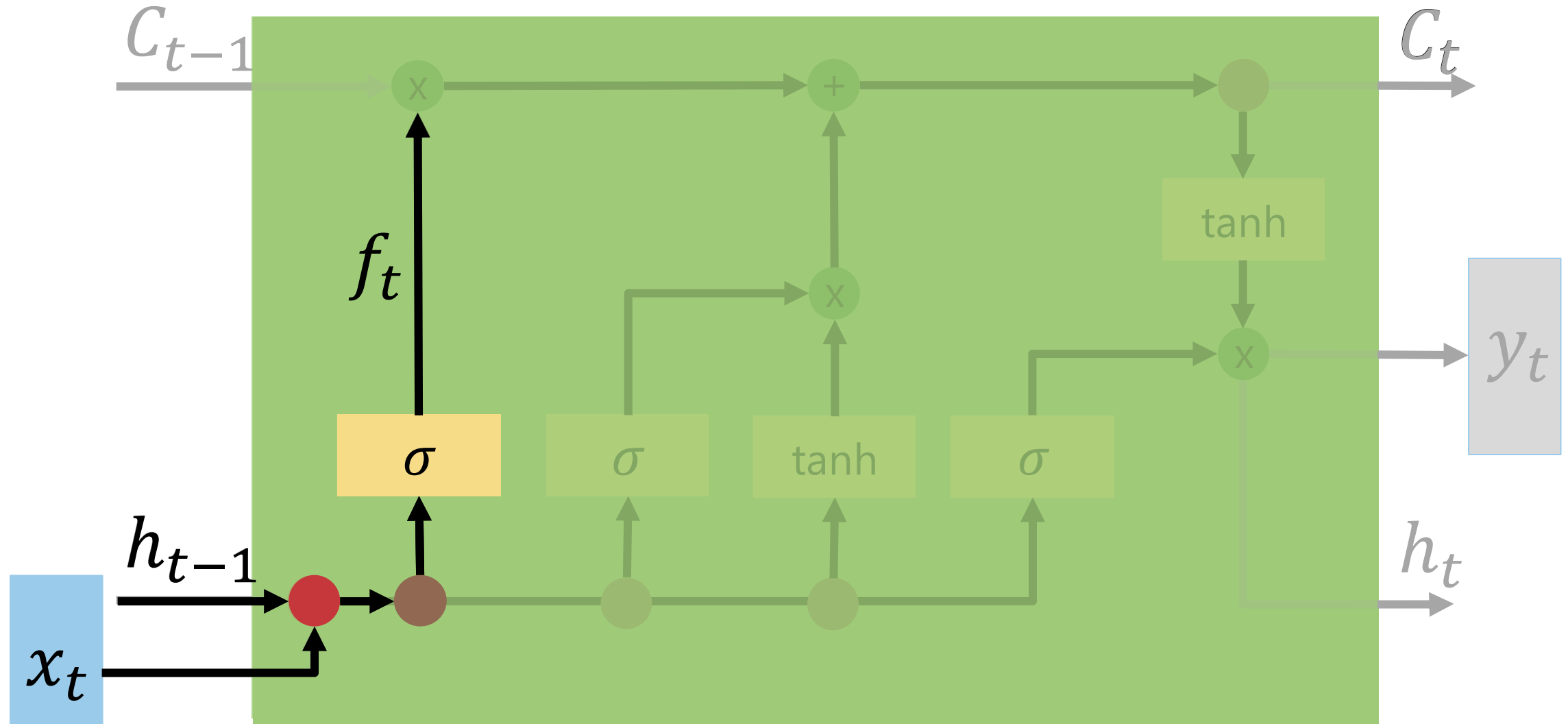
Cell State: the Key to the LSTM Structure



Gates Regulate Added & Removed Information

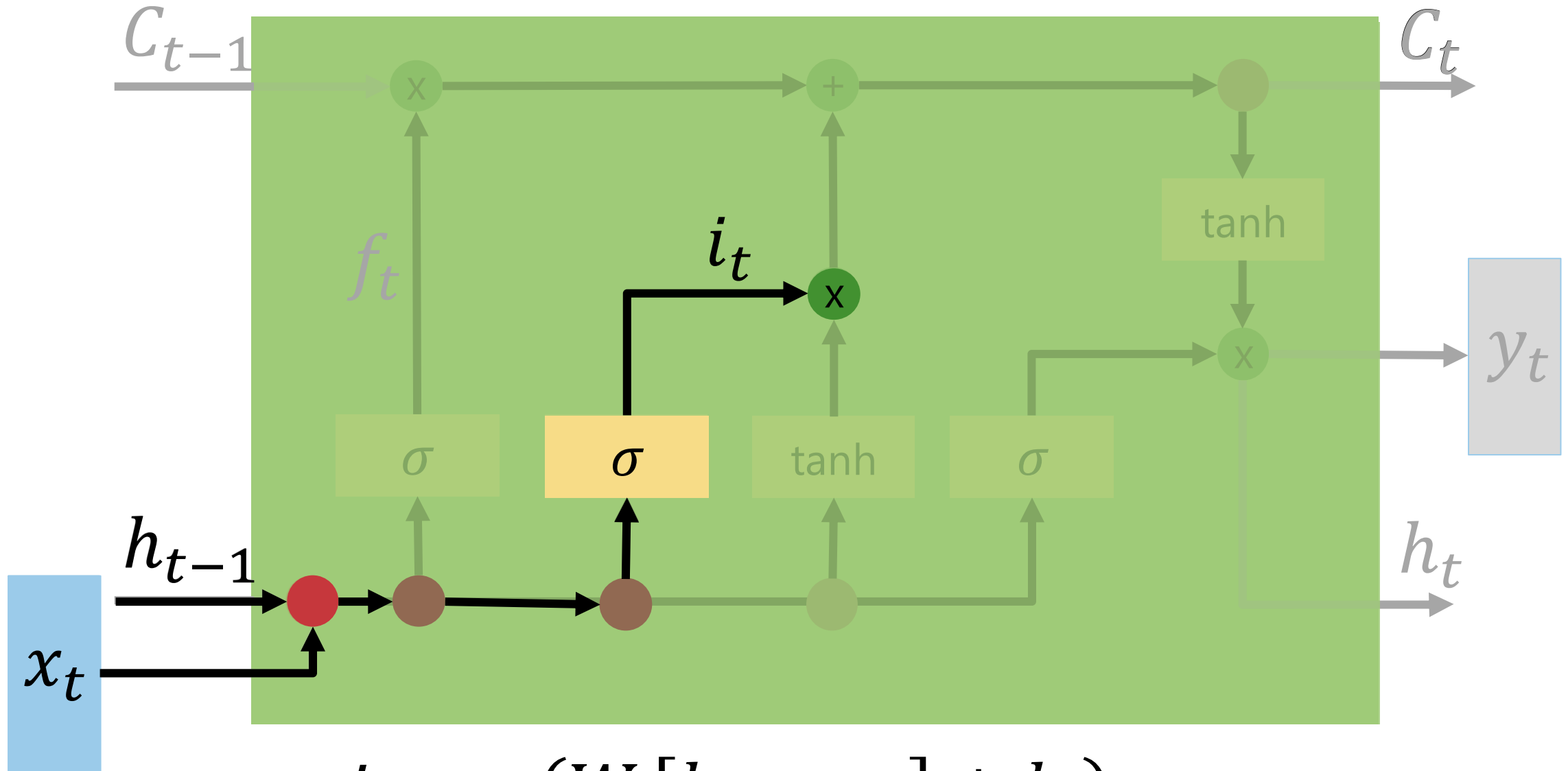


LSTM: Forget Gate Layer



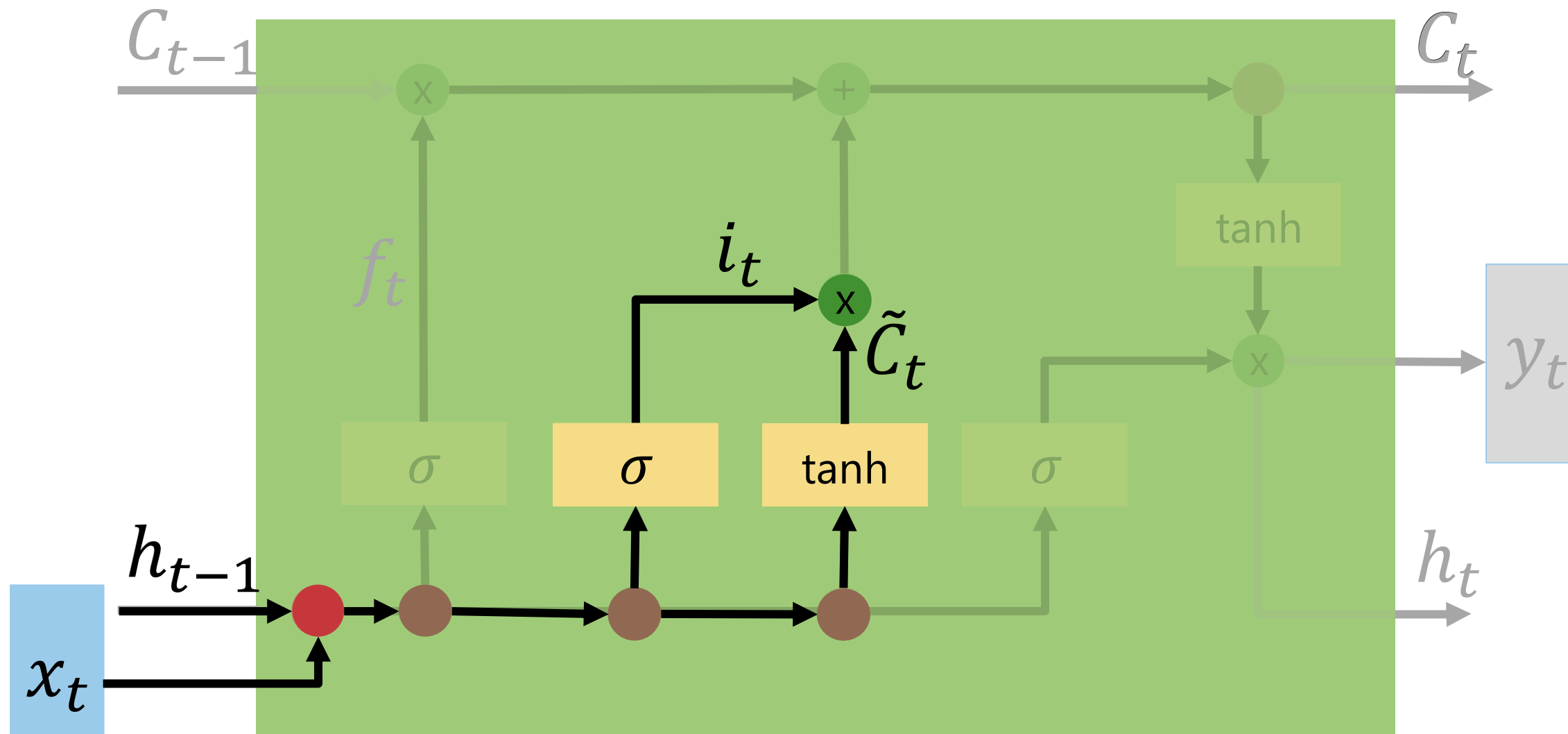
$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

LSTM: Input Gate Layer



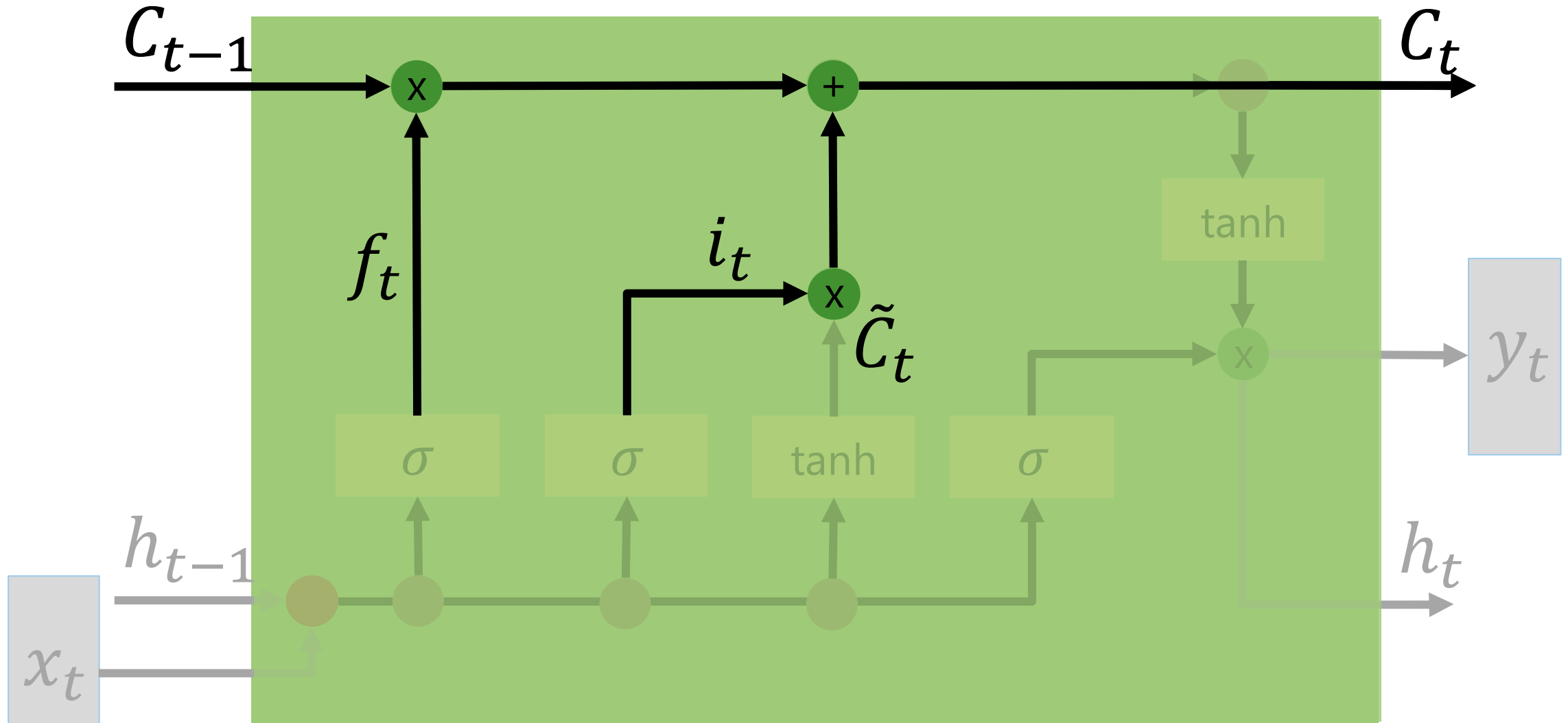
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

LSTM: New Candidate Values for the Cell State



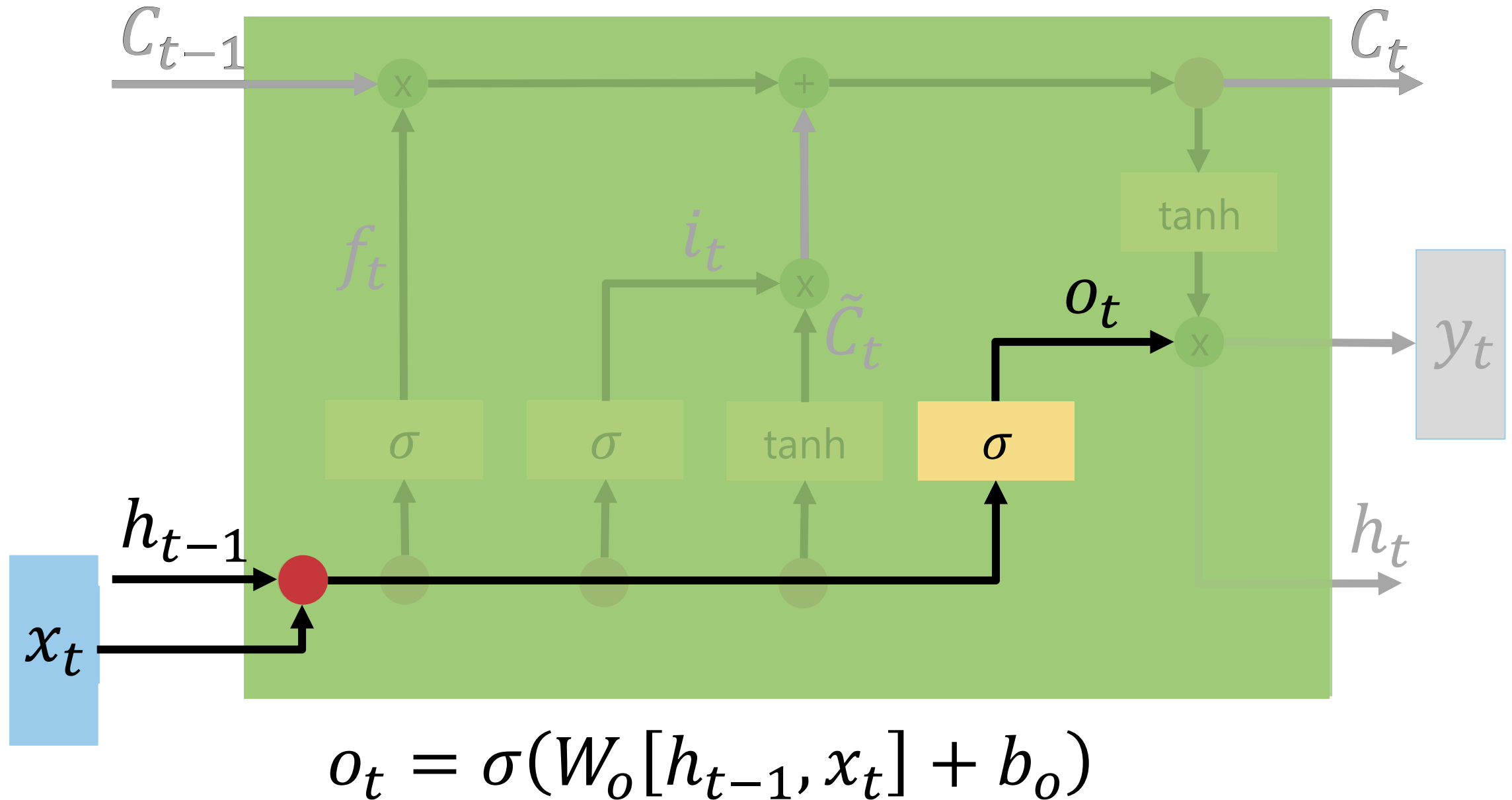
$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

Update the Cell State

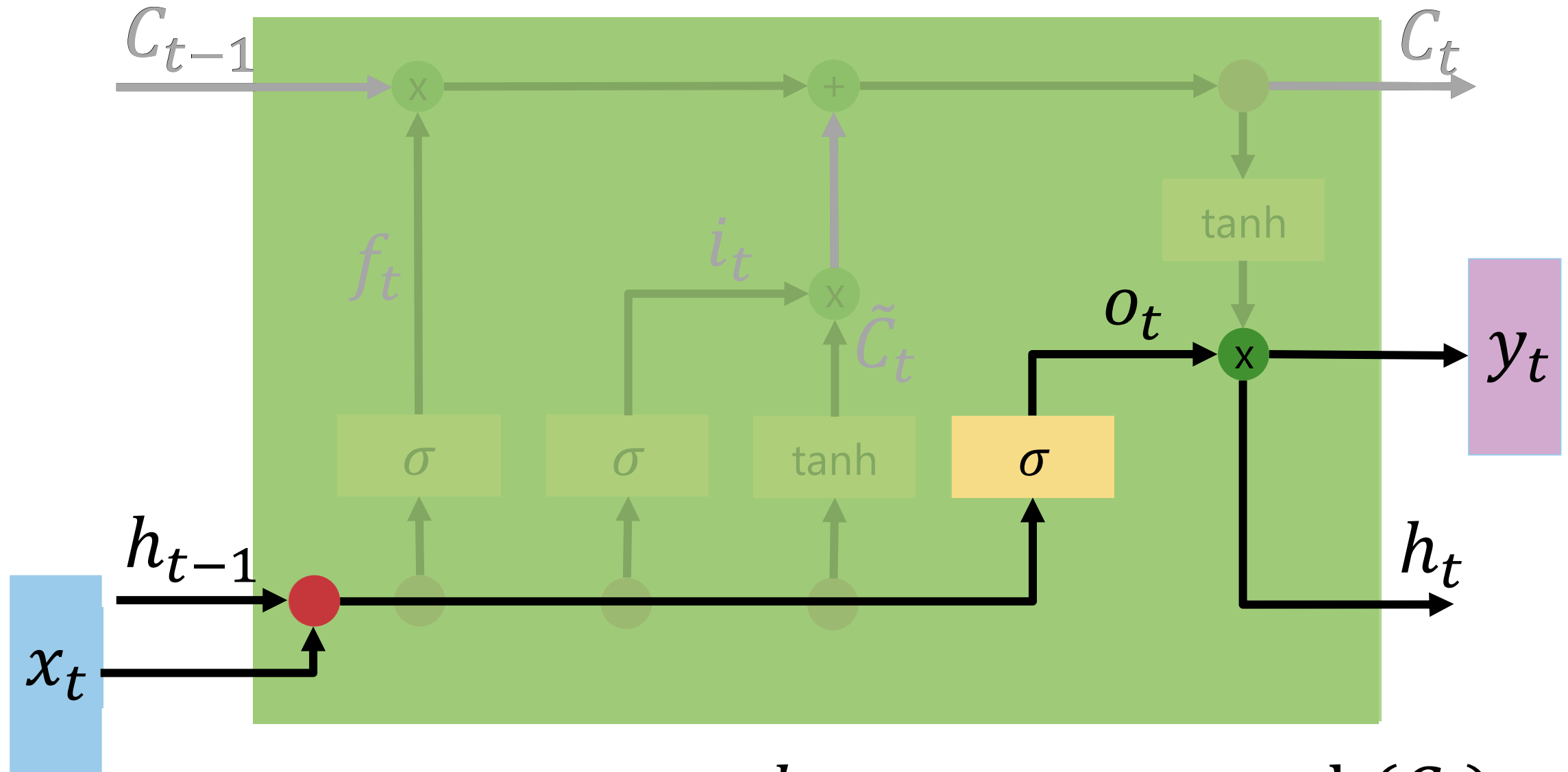


$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Gates Regulate Added & Removed Information

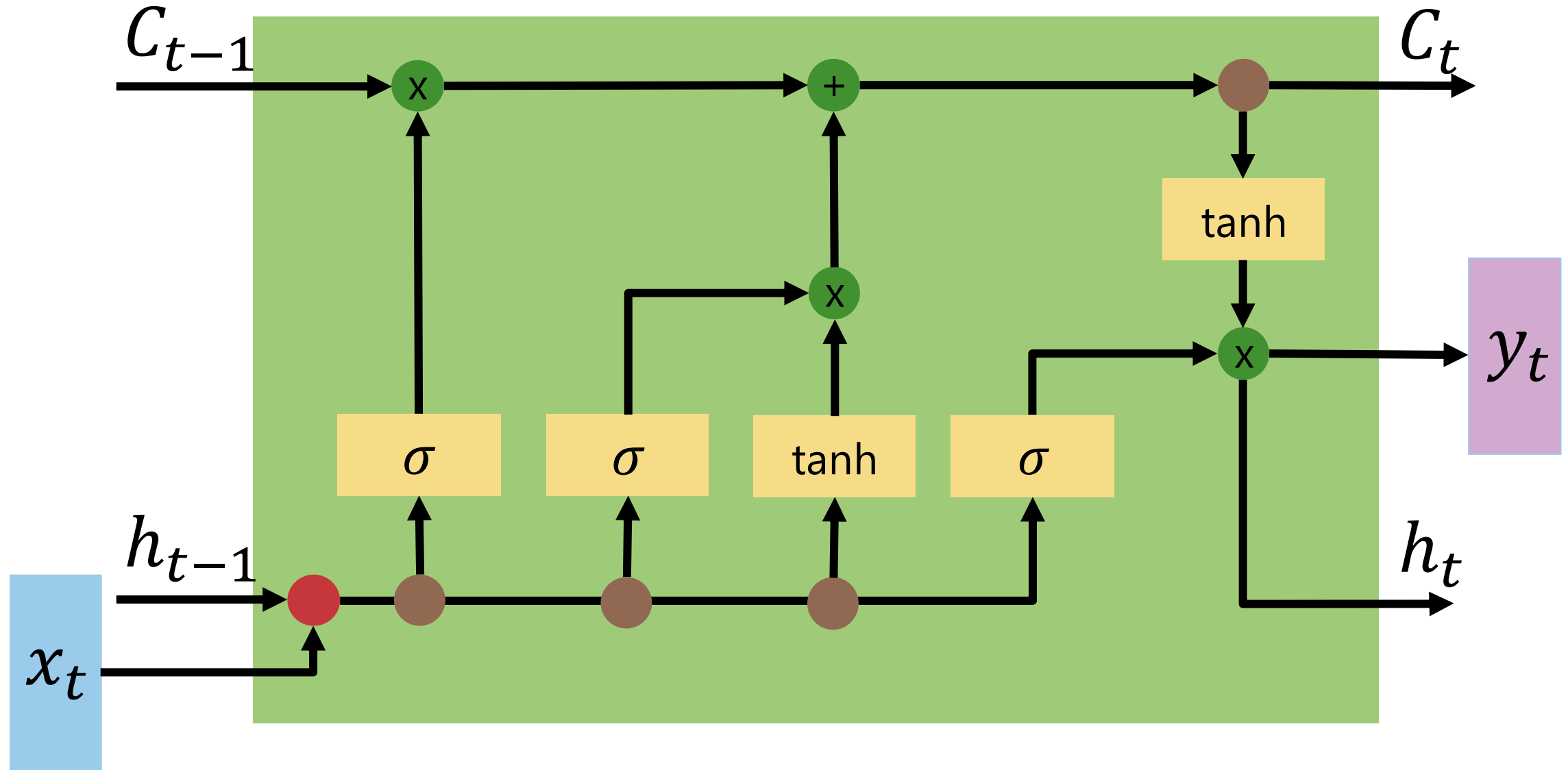


Gates Regulate Added & Removed Information

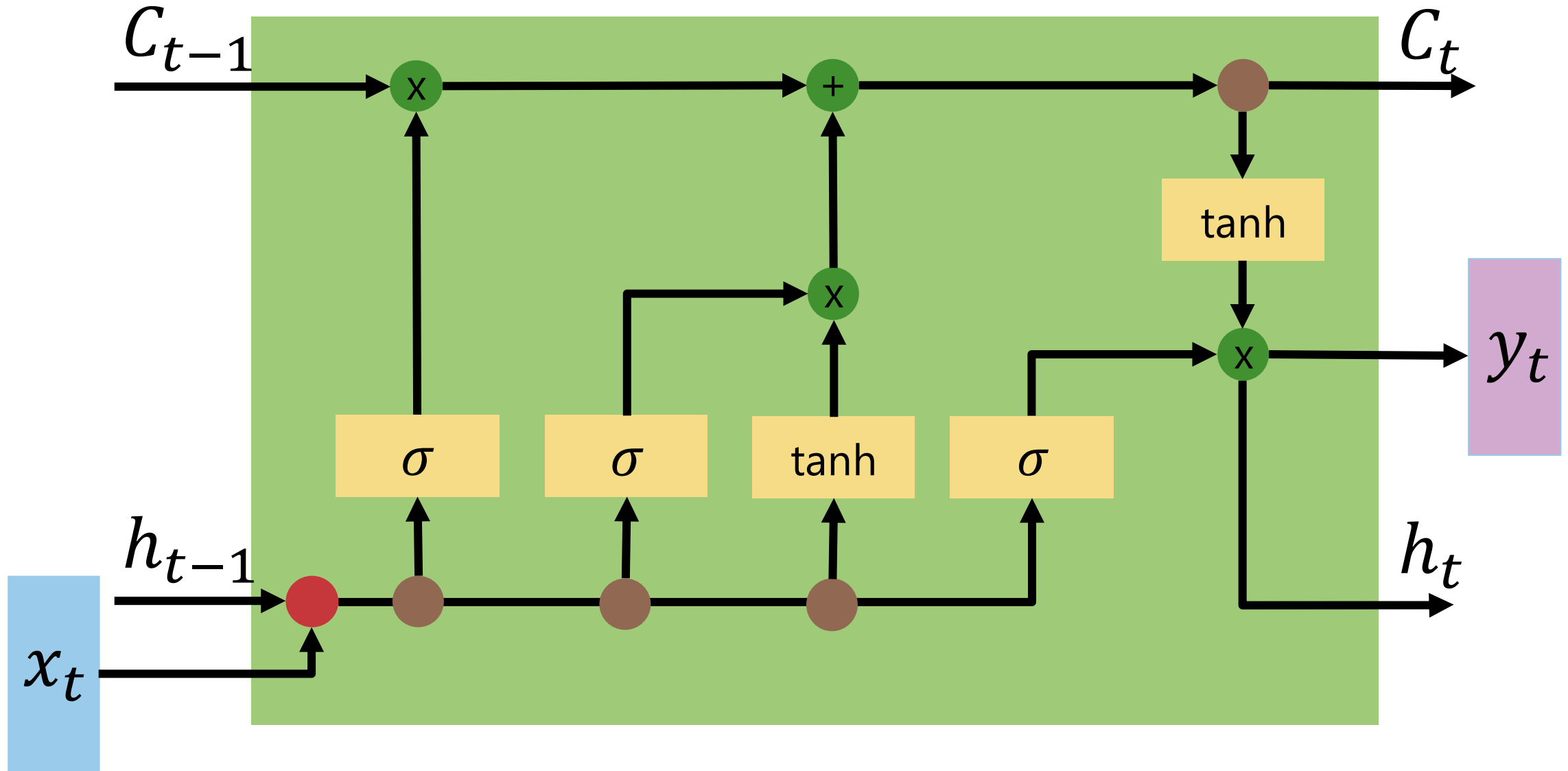


$$h_t = y_t = o_t * \tanh(C_t)$$

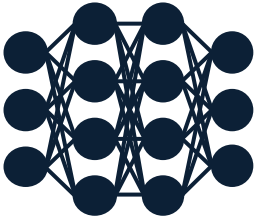
LSTM: Long Short-Term Memory



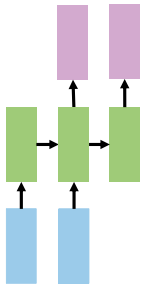
LSTM: Long Short-Term Memory



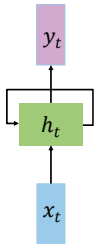
Summary



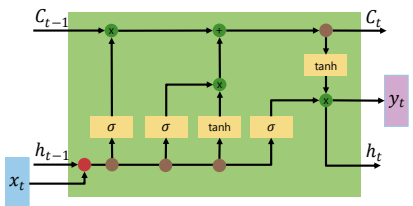
1. Standard neural networks have no memory.



2. Operate on sequence of vectors & remember the past



3. RNNs: Recurrent Neural Networks



4. LSTMs: Long Short-Term Memory Networks

Thank you!

Visit: <https://adam-dziedzic.com/lstms>

